Rosen — §8.1: Applications of Recurrence Relations

Selected Exercises

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Exercise 11

(a) Find a recurrence relation for the number of ways to climb n stairs if the person climbing the stairs can take one stair or two stairs at a time.

(b) What are the initial conditions?

(c) In how many ways can this person climb a flight of eight stairs?

Exercise 17

A string that contains only 0s, 1s, and 2s is called a ternary string.

(a) Find a recurrence relation for the number of ternary strings of length n that do not contain consecutive symbols that are the same.

(b) What are the initial conditions?

(c) How many ternary strings of length six do not contain consecutive symbols that are the same?

Exercise 18

(a) Find a recurrence relation for the number of ternary strings of length n that contain two consecutive symbols that are the same.

(b) What are the initial conditions?

(c) How many ternary strings of length six contain consecutive symbols that are the same?

Exercise 23

(a) Find the recurrence relation satisfied by S_n, where S_n is the number of regions into which three-dimensional space is divided by n planes if every three of the planes meet in one point, but no four of the planes go through the same point.

(b) Find S_n using iteration.

Exercise 26

(a) Find a recurrence relation for the number of ways to completely cover a $2 \times n$ checkerboard with 1×2 dominoes. [*Hint*: Consider separately the coverings where the position in the top right corner of the checkerboard is covered by a domino positioned horizontally and where it is covered by a domino positioned vertically.]

(b) What are the initial conditions for the recurrence relation in part (a)?

(c) How many ways are there to completely cover a 2×17 checkerboard with 1×2 dominoes?

Exercise 28

Show that the Fibonacci numbers satisfy the recurrence relation $f_n = 5f_{n-4} + 3f_{n-5}$ for $n = 5, 6, 7, \ldots$, together with the initial conditions $f_0 = 0$, $f_1 = 1$, $f_2 = 1$, $f_3 = 2$, and $f_4 = 3$. Use this recurrence relation to show that f_{5n} is divisible by 5, for $n = 1, 2, 3, \ldots$.

Exercise 29

Let $S(m, n)$ denote the number of onto functions from a set with m elements to a set with n elements. Show that $S(m, n)$ satisfies the recurrence relation

$$S(m,n) = n^m - \sum_{k=1}^{n-1} \binom{n}{k} S(m,k)$$

whenever $m \geq n$ and $n > 1$, with the initial condition $S(m, 1) = 1$.

Exercise 30

Recall (Example 5): C_n denotes the number of ways to parenthesize a product of $n+1$ numbers so as to determine the order of multiplication; $C_0 = 1$ and $C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$ for $n \geq 1$.

(a) Write out all the ways the product $x_0 \cdot x_1 \cdot x_2 \cdot x_3 \cdot x_4$ can be parenthesized to determine the order of multiplication.

(b) Use the recurrence relation developed in Example 5 to calculate C_4, the number of ways to parenthesize the product of five numbers so as to determine the order of multiplication. Verify that you listed the correct number of ways in part (a).

(c) Check your result in part (b) by finding C_4, using the closed formula for C_n mentioned in the solution of Example 5.

Exercise 32

In the Tower of Hanoi puzzle, suppose our goal is to transfer all n disks from peg 1 to peg 3, but we cannot move a disk directly between pegs 1 and 3. Each move of a disk must be a move involving peg 2. As usual, we cannot place a disk on top of a smaller disk.

(a) Find a recurrence relation for the number of moves required to solve the puzzle for n disks with this added restriction.

(b) Solve this recurrence relation to find a formula for the number of moves required to solve the puzzle for n disks.

(c) How many different arrangements are there of the n disks on three pegs so that no disk is on top of a smaller disk?

(d) Show that every allowable arrangement of the n disks occurs in the solution of this variation of the puzzle.

Exercise 35

Recall the Josephus problem: n people, numbered 1 to n, stand around a circle. In each stage, every second person still left alive is eliminated until only one survives. Denote the number of the survivor by $J(n)$.

Show that $J(n)$ satisfies the recurrence relation $J(2n) = 2J(n) - 1$ and $J(2n+1) = 2J(n) + 1$, for $n \geq 1$, and $J(1) = 1$.

Exercise 41

Recall the Reve's puzzle: the variation of the Tower of Hanoi puzzle with four pegs and n disks. The Frame–Stewart algorithm moves the disks from peg 1 to peg 4 by choosing an integer k with $1 \leq k \leq n$, recursively moving the $n-k$ smallest disks to peg 2 (using all four pegs), then moving the k largest disks to peg 4 (using the three-peg algorithm, without the peg holding the $n-k$ smallest disks), and finally recursively moving the $n-k$ smallest disks to peg 4.

Show that if $R(n)$ is the number of moves used by the Frame–Stewart algorithm to solve the Reve's puzzle with n disks, where k is chosen to be the smallest integer with $n \leq k(k+1)/2$, then $R(n)$ satisfies the recurrence relation $R(n) = 2R(n-k) + 2k - 1$, with $R(0) = 0$ and $R(1) = 1$.

Exercise 45

Show that $R(n)$ (the number of moves used by the Frame–Stewart algorithm, from Exercise 41) is $O(\sqrt{n}\, 2^{\sqrt{n}})$.

Exercise 52

Recall: the backward differences of a sequence $\{a_n\}$ are defined recursively by $\nabla a_n = a_n - a_{n-1}$ and $\nabla^{k+1} a_n = \nabla^k a_n - \nabla^k a_{n-1}$.

Show that any recurrence relation for the sequence $\{a_n\}$ can be written in terms of $a_n, \nabla a_n, \nabla^2 a_n, \ldots$. The resulting equation involving the sequence and its differences is called a *difference equation*.

Exercise 56

In this exercise we will develop a dynamic programming algorithm for finding the maximum sum of consecutive terms of a sequence of real numbers. That is, given a sequence of real numbers $a_1, a_2, \ldots, a_n$, the algorithm computes the maximum sum $\sum_{i=j}^{k} a_i$ where $1 \leq j \leq k \leq n$.

(a) Show that if all terms of the sequence are nonnegative, this problem is solved by taking the sum of all terms. Then, give an example where the maximum sum of consecutive terms is not the sum of all terms.

(b) Let $M(k)$ be the maximum of the sums of consecutive terms of the sequence ending at a_k. That is, $M(k) = \max_{1 \leq j \leq k} \sum_{i=j}^{k} a_i$. Explain why the recurrence relation $M(k) = \max(M(k-1) + a_k, a_k)$ holds for $k = 2, \ldots, n$.

(c) Use part (b) to develop a dynamic programming algorithm for solving this problem.

(d) Show each step your algorithm from part (c) uses to find the maximum sum of consecutive terms of the sequence $2, -3, 4, 1, -2, 3$.

(e) Show that the worst-case complexity in terms of the number of additions and comparisons of your algorithm from part (c) is linear.

Exercise 57

Dynamic programming can be used to develop an algorithm for solving the matrix-chain multiplication problem. This is the problem of determining how the product $A_1 A_2 \cdots A_n$ can be computed using the fewest integer multiplications, where $A_1, A_2, \ldots, A_n$ are $m_1 \times m_2$, $m_2 \times m_3$, $\ldots$, $m_n \times m_{n+1}$ matrices, respectively, and each matrix has integer entries. Recall that by the associative law, the product does not depend on the order in which the matrices are multiplied.

(a) Show that the brute-force method of determining the minimum number of integer multiplications needed to solve a matrix-chain multiplication problem has exponential worst-case complexity. [*Hint*: Do this by first showing that the order of multiplication of matrices is specified by parenthesizing the product, then use the Catalan-number count of parenthesizations (Exercise 30) together with its exponential lower bound.]

(b) Denote by A_{ij} the product $A_i A_{i+1} \cdots A_j$, and $M(i,j)$ the minimum number of integer multiplications required to find A_{ij}. Show that if the least number of integer multiplications are used to compute A_{ij}, where $i < j$, by splitting the product into the product of A_i through A_k and the product of A_{k+1} through A_j, then the first k terms must be parenthesized so that $A_{i,k}$ is computed in the optimal way using $M(i,k)$ integer multiplications and $A_{k+1,j}$ must be parenthesized so that $A_{k+1,j}$ is computed in the optimal way using $M(k+1,j)$ integer multiplications.

(c) Explain why part (b) leads to the recurrence relation

$$M(i,j) = \min_{i \le k < j} \big(M(i,k) + M(k+1,j) + m_i m_{k+1} m_{j+1} \big) \quad \text{if } 1 \le i < j \le n.$$

(d) Use the recurrence relation in part (c) to construct an efficient algorithm for determining the order the n matrices should be multiplied to use the minimum number of integer multiplications. Store the partial results $M(i,j)$ as you find them so that your algorithm will not have exponential complexity.

(e) Show that your algorithm from part (d) has $O(n^3)$ worst-case complexity in terms of multiplications of integers.